

Human Development Index (HDI) Predictor

Project Planning

Project: Human Development Index (HDI) Predictor
Phase: Project Planning Phase
Developer: Nainaru Thulasi Ram

1. Epics Overview

Work was organized into 8 epics, matching the machine learning lifecycle from environment setup through to the deployed Flask web application.

2. Epic Breakdown

Epic	Description	Duration
1. Environment Setup & Package Installation	Install Python libraries; create the project folder structure	2h
2. Importing Required Libraries	Import NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, Flask	3h
3. Dataset Download & Understanding	Source and load the HDI dataset; inspect structure	3h
4. Data Preprocessing & Label Encoding	Select variables, handle nulls, encode Country	3h
5. Train/Test Split	Split data 90/10 for training and evaluation	1h
6. Fitting the Model	Train Linear Regression; predict; evaluate with R ² /MAE/RMSE	2h
7. Saving the Model	Serialize model and encoder with Pickle	1h
8. Building the Flask Web Application	Backend routes, HTML templates, end-to-end testing	1.5h

3. Individual Ownership

This project was completed individually by Nainaru Thulasi Ram, who was responsible for every stage of the pipeline — dataset preparation and EDA, preprocessing and label encoding, model training and evaluation, and the Flask backend and frontend.

4. Milestones

- Milestone 1: Dataset generated and EDA completed
- Milestone 2: Model trained and evaluated ($R^2 \approx 0.99$)
- Milestone 3: Flask app functional end-to-end with live predictions
- Milestone 4: Documentation completed and project packaged for submission